

Research Notes on Cross-validation statistics

Cross-validation statistics

>> Section 1

General case In most other regression procedures (e.g. logistic regression), there is no simple formula to compute the expected out-of-sample fit. Cross-validation is, thus, a generally applicable way to predict the performance of a model on unavailable data using numerical computation in place of theoretical analysis.

>> Section 2

Motivation Assume a model with one or more unknown parameters, and a data set to which the model can be fit (the training data set). The fitting process optimizes the model parameters to make the model fit the training data as well as possible. If an independent sample of validation data is taken from the same population as the training data, it will generally turn out that the model does not fit the validation data as well as it fits the training data. The size of this difference is likely to be large especially when the size of the training data set is small, or when the number of parameters in the model is large. Cross-validation is a way to estimate the size of this effect.

>> Section 3

By performing an initial analysis to identify the most informative features using the entire data set – if feature selection or model tuning is required by the modeling procedure, this must be repeated on every training set. Otherwise, predictions will certainly be upwardly biased. If cross-validation is used to decide which features to use, an inner cross-validation to carry out the feature selection on every training set must be performed. Performing mean-centering, rescaling, dimensionality reduction, outlier removal or any other data-dependent preprocessing using the entire data set. While very common in practice, this has been shown to introduce biases into the cross-validation estimates. By allowing some of the training data to also be included in the test set – this can happen due to "twinning" in the data set, whereby some exactly identical or nearly identical samples are present in the data set, see pseudoreplication. To some extent twinning always takes place even in perfectly independent training and validation samples. This is because some of the training sample observations will have nearly identical values of predictors as validation sample observations. And some of these will correlate with a target at better than chance levels in the same direction in both training and validation when they are actually driven by confounded predictors with poor external validity. If such a cross-validated model is selected from a k-fold set, human confirmation bias will be at work and determine that such a model has been validated. This is why traditional cross-validation needs to be supplemented with controls for human bias and

confounded model specification like swap sampling and prospective studies.

>> Section 4

Leave-one-out cross-validation (LOOCV) is a particular case of leave-p-out cross-validation with $p = 1$. The process looks similar to jackknife; however, with cross-validation one computes a statistic on the left-out sample(s), while with jackknifing one computes a statistic from the kept samples only. LOO cross-validation requires less computation time than LpO cross-validation because there are only $C_1^n = n$ $\{\displaystyle C_{\{1\}}^n = n\}$ passes rather than C_p^n $\{\displaystyle C_{\{p\}}^n\}$. However, n $\{\displaystyle n\}$ passes may still require quite a large computation time, in which case other approaches such as k-fold cross validation may be more appropriate. Pseudo-code algorithm: Input: x , {vector of length N with x -values of incoming points} y , {vector of length N with y -values of the expected result} interpolate(x_{in} , y_{in} , x_{out}), { returns the estimation for point x_{out} after the model is trained with x_{in} - y_{in} pairs} Output: err, {estimate for the prediction error} Steps:

>> Section 5

Computational issues Most forms of cross-validation are straightforward to implement as long as an implementation of the prediction method being studied is available. In particular, the prediction method can be a "black box" – there is no need to have access to the internals of its implementation. If the prediction method is expensive to train, cross-validation can be very slow since the training must be carried out repeatedly. In some cases such as least squares and kernel regression, cross-validation can be sped up significantly by pre-computing certain values that are needed repeatedly in the training, or by using fast "updating rules" such as the Sherman–Morrison formula. However one must be careful to preserve the "total blinding" of the validation set from the training procedure, otherwise bias may result. An extreme example of accelerating cross-validation occurs in linear regression, where the results of cross-validation have a closed-form expression known as the prediction residual error sum of squares (PRESS).

>> Section 6

Measures of fit The goal of cross-validation is to estimate the expected level of fit of a model to a data set that is independent of the data that were used to train the model. It can be used to estimate any quantitative measure of fit that is appropriate for the data and model. For example, for binary classification problems, each case in the validation set is either predicted correctly or incorrectly. In this situation the misclassification error rate can be used to

Research Notes on Cross-validation statistics

Cross-validation statistics

summarize the fit, although other measures derived from information (e.g., counts, frequency) contained within a contingency table or confusion matrix could also be used. When the value being predicted is continuously distributed, the mean squared error, root mean squared error or median absolute deviation could be used to summarize the errors.

>> Section 7

Hoornweg (2018) shows that a loss function with such an accuracy-simplicity tradeoff can also be used to intuitively define shrinkage estimators like the (adaptive) lasso and Bayesian / ridge regression. Click on the lasso for an example.

>> Section 8

Applications Cross-validation can be used to compare the performances of different predictive modeling procedures. For example, suppose we are interested in optical character recognition, and we are considering using either a Support Vector Machine (SVM) or k-nearest neighbors (KNN) to predict the true character from an image of a handwritten character. Using cross-validation, we can obtain empirical estimates comparing these two methods in terms of their respective fractions of misclassified characters. In contrast, the in-sample estimate will not represent the quantity of interest (i.e. the generalization error). Cross-validation can also be used in variable selection. Suppose we are using the expression levels of 20 proteins to predict whether a cancer patient will respond to a drug. A practical goal would be to determine which subset of the 20 features should be used to produce the best predictive model. For most modeling procedures, if we compare feature subsets using the in-sample error rates, the best performance will occur when all 20 features are used. However under cross-validation, the model with the best fit will generally include only a subset of the features that are deemed truly informative. A recent development in medical statistics is its use in meta-analysis. It forms the basis of the validation statistic, V_n which is used to test the statistical validity of meta-analysis summary estimates. It has also been used in a more conventional sense in meta-analysis to estimate the likely prediction error of meta-analysis results.